

Supplementary Analysis: When to Intervene

Anonymous submission 2613

1 Outcomes and paired comparisons

The analyses use all 24 participants and 216 rounds (72 per condition). Every participant completed three rounds per condition, and all 24 contributed final-survey ratings. C denotes control, K constant assistance, and A adaptive assistance.

The four main outcomes are completion, correct pieces, adapted workload, and frustration. For each outcome, two-sided paired t tests compare participants' three-round condition averages across K-C, A-C, and A-K. Holm correction jointly covers these twelve tests. A separate nineteen-test group covers three contrasts each for mental demand, physical demand, temporal demand, performance, and effort, and A-K contrasts for four assistance-experience items. These analyses and comparison groups are exploratory, not preregistered. Recorded round duration and final-survey ratings are descriptive.

Correct pieces range from 0 to 7, using the admissible orientation with the highest correct count. Completion corresponds to a solved seven-piece arrangement. Workload averages six seven-point burden-aligned items, reversing performance as $8 - P$, then rescales the average B to $(B - 1)100/6$. Frustration is both a main outcome and a component of this composite; the two effects are not independent replications. Completion and correct pieces are also linked.

Table 1 reports pointwise confidence intervals, not simultaneous multiplicity-adjusted intervals. p_H denotes the Holm-adjusted value. Cohen's d_z divides the paired mean difference by the standard deviation of participant differences. Significance is evaluated using $p_H < .05$, not whether an unadjusted interval excludes zero.

Table 1: Main paired contrasts ($n = 24$); completion differences are percentage points. Twelve-test Holm group.

Outcome	Contrast	Difference	95% CI	d_z	p_H
Completion	K-C	44.44	[32.23, 56.66]	1.54	< .001
Completion	A-C	34.72	[21.94, 47.50]	1.15	< .001
Completion	A-K	-9.72	[-23.16, 3.71]	-0.31	0.444
Correct pieces	K-C	2.25	[1.61, 2.89]	1.47	< .001
Correct pieces	A-C	2.00	[1.46, 2.54]	1.56	< .001
Correct pieces	A-K	-0.25	[-0.71, 0.21]	-0.23	0.532
Workload	K-C	-12.96	[-17.40, -8.52]	-1.23	< .001
Workload	A-C	-16.40	[-21.95, -10.84]	-1.25	< .001
Workload	A-K	-3.43	[-9.66, 2.80]	-0.23	0.532
Frustration	K-C	-0.76	[-1.28, -0.25]	-0.63	0.027
Frustration	A-C	-1.18	[-1.70, -0.66]	-0.96	< .001
Frustration	A-K	-0.42	[-0.90, 0.07]	-0.36	0.363

2 Puzzle- and period-adjusted robustness

All adjusted models include condition, categorical puzzle identity, and categorical block period. Completion uses a binomial-logit generalized estimating equation with independent working correlation and participant-clustered Mancl-DeRouen bias-reduced covariance. The working correlation does not treat a person's rounds as independent. Correct pieces, workload, and numeric ratings use participant-fixed-effect ordinary least squares with CR1 participant-clustered covariance. CR1 applies the finite-sample factor $G/(G - 1)(N - 1)/(N - k)$, where G is the participant count, N the round count, and k the parameter count. Contrasts use a t reference with 23 degrees of freedom.

Completion contrasts are odds ratios; their tests use log-odds coefficients. Other adjusted effects are mean-score differences. They are not interchangeable with the observed paired differences. The twelve/nineteen correction groups remain the same. The adjusted models support the same main contrast conclusions: both assistance conditions outperform control on completion and pieces and reduce workload and frustration; none establishes an A-K difference. Constant-control frustration has adjusted $p_H = .045$, compared with .027 in the paired analysis.

These are supplementary robustness checks, not a validation of the HR-rise cue or evidence of equivalence. With 24 participant clusters, covariance corrections and the t reference remain approximations. Completion models converged with full-rank designs; base fitted probabilities ranged from .053 to .805. Numeric fixed-effect models can predict beyond bounded scales (14 piece predictions outside 0-7; one workload prediction outside 0-100); predictions were not clipped. A further categorical within-block-position check retained the direction of the main contrasts; its unadjusted diagnostic p values are reported separately and do not replace the main tests.

Table 2: Puzzle/block-adjusted main contrasts. Completion uses odds ratios (OR); other effects are score differences. Twelve-test Holm group.

Outcome	Contrast	Effect	95% CI	p_H
Completion (OR)	K-C	8.16	[3.84, 17.33]	< .001
Completion (OR)	A-C	5.43	[2.41, 12.24]	0.002
Completion (OR)	A-K	0.67	[0.34, 1.30]	0.802
Correct pieces	K-C	2.14	[1.54, 2.74]	< .001
Correct pieces	A-C	1.95	[1.41, 2.48]	< .001
Correct pieces	A-K	-0.19	[-0.74, 0.35]	0.802
Workload	K-C	-12.15	[-16.94, -7.37]	< .001
Workload	A-C	-15.04	[-20.76, -9.31]	< .001
Workload	A-K	-2.88	[-9.38, 3.61]	0.802
Frustration	K-C	-0.72	[-1.24, -0.20]	0.045
Frustration	A-C	-1.04	[-1.58, -0.50]	0.003
Frustration	A-K	-0.32	[-0.83, 0.18]	0.802

Table 3: Additional within-block-position sensitivity: unadjusted diagnostic p values, not replacements for family-adjusted tests.

Outcome	K-C	A-C	A-K
Completion	< .001	< .001	0.202
Correct pieces	< .001	< .001	0.449
Workload	< .001	< .001	0.353
Frustration	0.008	< .001	0.181

3 Secondary ratings and study-wide experience

Table 4 provides every secondary contrast under the nineteen-test correction. Assisted-experience comparisons use 144 assisted rounds for adjusted models and all 24 participant averages for paired tests. Final helpfulness, efficacy, trust, and reliance summarize the study as a whole, not separate policies. Their means (SDs) are 5.00 (1.53), 4.83 (1.55), 4.79 (1.59), and 4.88 (1.48), respectively.

Table 4: Secondary ratings: nineteen-test Holm group, separate from the twelve main contrasts.

Rating	Contrast	Difference	95% CI	Paired p_H	Model p_H
Helpfulness	A-K	0.25	[-0.16, 0.66]	1.000	1.000
Timing	A-K	0.56	[0.16, 0.95]	0.103	0.210
Seamlessness	A-K	0.08	[-0.45, 0.61]	1.000	1.000
Frustration relief	A-K	0.40	[-0.02, 0.83]	0.623	0.906
Mental demand	K-C	-0.89	[-1.24, -0.54]	< .001	0.001
Mental demand	A-C	-1.08	[-1.50, -0.67]	< .001	< .001
Mental demand	A-K	-0.19	[-0.70, 0.32]	1.000	1.000
Physical demand	K-C	-0.38	[-0.70, -0.05]	0.320	0.717
Physical demand	A-C	-0.60	[-1.13, -0.07]	0.320	0.736
Physical demand	A-K	-0.22	[-0.69, 0.24]	1.000	1.000
Temporal demand	K-C	-0.29	[-0.64, 0.06]	0.867	1.000
Temporal demand	A-C	-0.36	[-0.84, 0.12]	1.000	1.000
Temporal demand	A-K	-0.07	[-0.66, 0.52]	1.000	1.000
Performance	K-C	1.61	[1.05, 2.17]	< .001	< .001
Performance	A-C	2.01	[1.42, 2.61]	< .001	< .001
Performance	A-K	0.40	[-0.18, 0.98]	1.000	1.000
Effort	K-C	-0.74	[-1.08, -0.39]	0.003	0.015
Effort	A-C	-0.67	[-1.11, -0.23]	0.064	0.177
Effort	A-K	0.07	[-0.27, 0.41]	1.000	1.000

4 Condition orders and puzzle allocation

Four participants followed each of CKA, CAK, KCA, KAC, ACK, and AKC. Every participant attempted all nine distinct puzzles. Order balance did not imply equal puzzle allocation by condition; Table 5 shows the realized assignment.

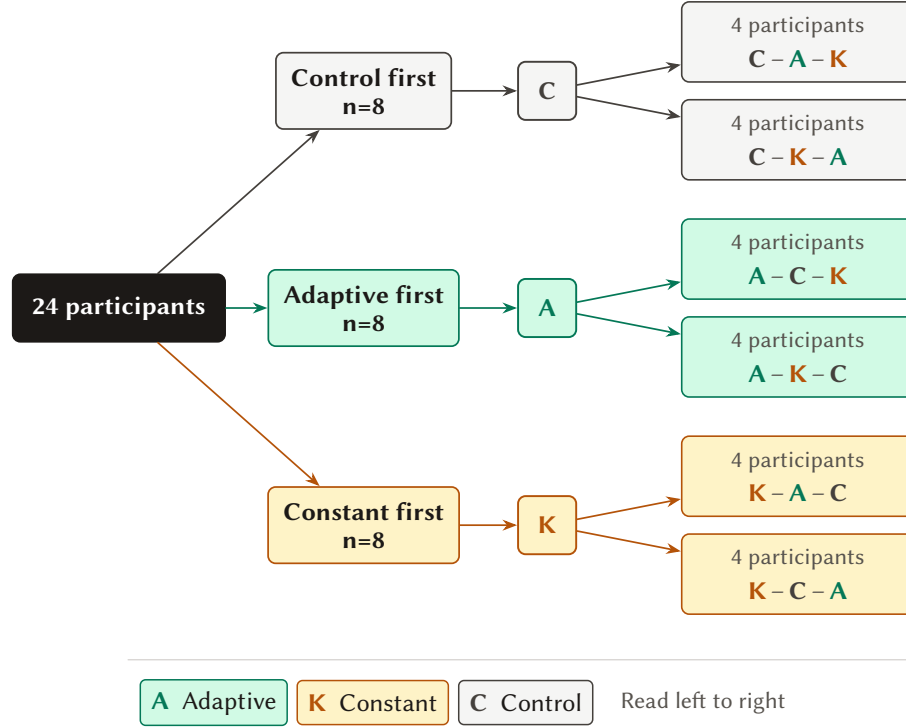


Figure 1: Condition orders: four participants per order, 24 total.

Table 5: Puzzle allocation; each condition has 72 attempts.

Puzzle	Control	Constant	Adaptive	Total
1	8	11	5	24
2	9	10	5	24
3	9	12	3	24
4	8	7	9	24
5	5	8	11	24
6	6	3	15	24
7	6	8	10	24
8	11	7	6	24
9	10	6	8	24

5 Reproducibility and interpretation

The analysis used Python 3.12.14, NumPy 2.3.3, SciPy 1.16.2, pandas 2.3.3, statsmodels 0.14.5, and patsy 1.0.1. Independent checks reproduced paired tests, cluster-covariance calculations, and Holm adjustment. No rounds were excluded or imputed in these analyses. Aggregate comparison CSVs accompany this supplement; raw participant files and recordings are not included.

The policies share an available preset-hint list and retrieval rule, not a demonstrated equal intervention dose. Effects therefore concern the assistance packages, not timing alone or robot retrieval in isolation. Recorded duration reflects round end-marking for solved and unsolved attempts and is not treated as verified active solving time. Nonsignificance is not evidence that two policies are equivalent.